A1.1 Water

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a1-1.html

Written for this site from the class slides. The chapter is on pages 2–10 of the course textbook.
Image credits are on the last slide of the slides for this subtopic.

A1.1.1 Water is the medium for life

pp. 2–10

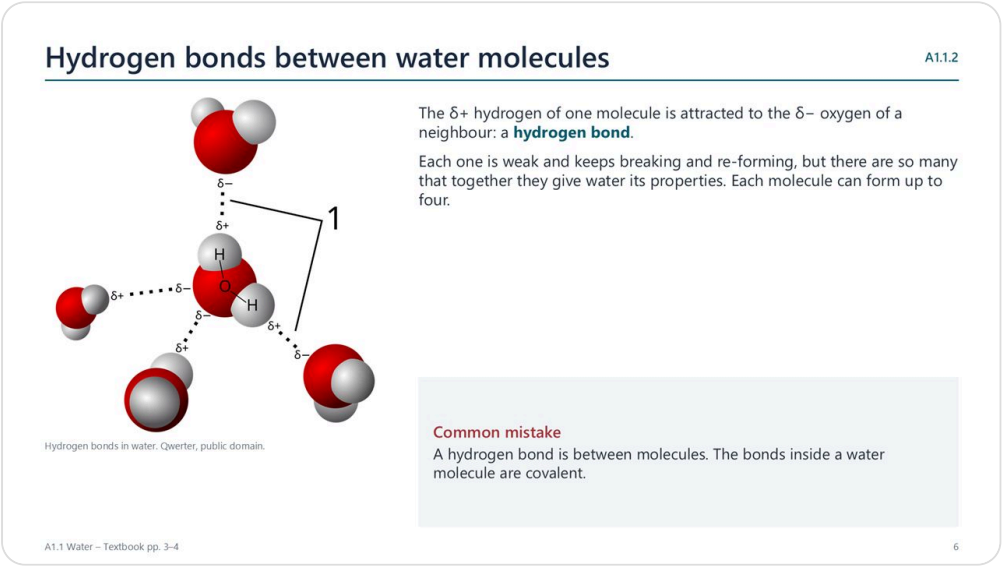
The first cells arose in water, and most processes of life still occur in it. Water makes up much of the cytoplasm, and it is the main part of blood plasma.

A1.1.2 Polar covalent bonds and hydrogen bonds

pp. 2–10

Oxygen attracts the shared electrons in its covalent bonds more strongly than hydrogen does, so oxygen has a slight negative charge, and hydrogen a slight positive charge. The molecule is **polar**.

The slightly positive hydrogen of one molecule attracts the slightly negative oxygen of another. This is a **hydrogen bond**. One is weak, but there are very many in water.



Hydrogen bonds in water. Qwerter, public domain.

A1.1.3 Cohesion

pp. 2–10

Cohesion is the attraction between water molecules, due to hydrogen bonds. It keeps a column of water intact under tension in **xylem**. It also causes **surface tension**, so a water surface can be a habitat for small animals such as water striders.

A1.1.4 Adhesion

pp. 2–10

Adhesion is the attraction between water and polar or charged materials. It causes **capillary action**, which draws water up through narrow channels in soil, and through cellulose in plant cell walls.

Watch out: cohesion is between water molecules, and adhesion is between water and something else.

A1.1.5 Water as a solvent

pp. 2–10

A wide variety of **hydrophilic** substances dissolve in water, and most enzymes catalyse reactions in aqueous solution. Water also transports substances in blood, xylem and phloem.

Substance	Solubility	How it is carried in blood
Glucose	Soluble	Dissolved in plasma
Sodium chloride	Soluble	Dissolved in plasma, as ions
Oxygen	Poorly soluble	Mostly bound to haemoglobin
Fats and cholesterol	Insoluble	Inside lipoprotein particles

Some molecules depend on being **hydrophobic**. For example, the core of a membrane forms a barrier, and leaf wax reduces water loss.

A1.1.6 Physical properties and aquatic animals

pp. 2–10

Property	Water compared with air	Ringed seal	Black-throated loon
Buoyancy	Much higher	Fat helps it float	Dense bones, and air sacs it can compress to dive
Viscosity	Much higher: more drag	Streamlined shape, flippers	Torpedo shape, webbed feet
Thermal conductivity	Much higher: heat lost fast	Thick blubber	Dense, oiled feathers
Specific heat	Much higher: stable temperature	A stable habitat for both	

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